

Task 1

RSAD2_MOUSE	1	MGMLVPTALAARLLSLFQQQLGSLWSGLAILFCWLRIALGWLDPGKEQPQ	50
G9MQB8_HYPVG	1	-----	0
RSAD2_MOUSE	51	VRGEPEDTQETQEDGNSTQP-TTPVSVNYHFTRQCNKCGFCFHTAKTSF	99
G9MQB8_HYPVG	1	----MKDNQEAAAFGGRVRTGQVPVSVNYHFSRKCNKECLFCFHTATTSH	46
RSAD2_MOUSE	100	VLPLEEAKRGLLLLKQAGLEKINFSGGEPFLQDRGEYLGKLVRFCKEELA	149
G9MQB8_HYPVG	47	VEKPENAKRGLTLLKQAGMKINFAGGEPFLYPK--FLGEMIDFCKETLQ	94
RSAD2_MOUSE	150	LPSVSIVSNGSLIRERWFKDYGEYLDILAISCDSFDEQVNALIGRGQGKK	199
G9MQB8_HYPVG	95	LESVSIVTNGSLVKEQFLQKHGRNIDILAVSCDSFNEATNIKIGRSG--	142
RSAD2_MOUSE	200	NHVENLQKLRRWCRDYKVAFKINSVINRFNVDDEMNEHIKALSPVRWKVF	249
G9MQB8_HYPVG	143	DNVQKLYEIGSWCQKYDIKFLNTVVNKFNHLEDMNDHLNALQPFRWKCF	192
RSAD2_MOUSE	250	QCLLIEGENSGEDALREAERFLISNEEFETFLERHKEVSCLVPESNQKMK	299
G9MQB8_HYPVG	193	QVLIVTGENSDKTLRNAHSLTISDDEFDRFCERHSSQTCLVPEPNRLMA	242
RSAD2_MOUSE	300	DSYLILDEYMRFLNCTGGRKDPKSIIDVGVEEAIKFSGFDEKMFLKRGG	349
G9MQB8_HYPVG	243	KSYLILDEYMRFLDRNG--QQPSKSILEVGVQQALQAVFWDEEAFVERGG	290
RSAD2_MOUSE	350	KYVWSK----ADLK-LDW	362
G9MQB8_HYPVG	291	IYDWNKSSCSDSKDLEW	308

Name	Motif	Functional role	Comment
SAM-binding motif	KINFSGGEPF	Binding of SAM	
ddh-synthase reaction motif	NΦHXXXCXXXCXXCF	- CX3CX2CF interact with a [4Fe-4S] cluster located at the base of the catalytic site - plays role in ddhNTP formation	Φ represents either Trp, Tyr, or Phe, X represents any amino acid

Q1

The main difference between the aligned proteins is the long end gap at the N-terminal sequence of fungi protein.

Q2

Active site motifs are annotated in the sequence with orange and blue color accordingly. These sequences are highly conserved, with the exception of just one amino acid in the SAM-binding region, indicating that they are functional.

It is noticeable that the N-terminal tail of mouse protein consists of mostly hydrophobic amino acids, such as: A, V, I, L, M, F, W. It follows that this mouse protein is probably attached to the membrane. While fungus protein is soluble. Other than localisation difference, these proteins are most likely catalytically active as a radical-SAM viperin, likely able to make ddhNTP.

Task 2

Q1

BLOSUM type	Identity	Similarity	SAM-binding motif	ddh-synthase motif	reaction
BLOSUM62	100/403 (24.8%)	145/403 (36.0%)	KINFSGGEPF :: : . RLNFAGGEPM	NYHFTRQCNYKCGFCF ::. NWHITEACNYNCSYCF	
BLOSUM45	107/409 (26.2%)	151/409 (36.9%)	KINFSGGEPF :: : . RLNFAGGEPM	NYHFTRQCNYKCGFCF ::. NWHITEACNYNCSYCF	
BLOSUM30	102/404 (25.2%)	165/404 (40.8%)	KI--NFSGGEPF .: : . SVRLNFAGGEPM	NYHFTRQCNYKCGFCF : . .:. . .:. NWHITEACNYNCSYCF	

Q2

1. What type of matrix is the best for the analyzed proteins?

BLOSUM62: Motifs clearly seen.

BLOSUM45: Slightly higher identity than BLOSUM62. Motifs remain cleanly aligned.

BLOSUM30: Identity is lower. Increases overall similarity but also allows more questionable matches/gaps. Starts to distort the alignment (small gaps near the SAM motif), which suggests over-permissive scoring.

For these distant homologs, BLOSUM45 is the most appropriate choice. It slightly improves identity over BLOSUM62 and preserves the SAM-binding and ddh-synthase motifs clearly,

without over-smoothing the alignment. BLOSUM30 is too permissive for reliable motif/functional interpretation.

2. What are the other parameters that may influence the result?

- **Gap penalties** (gap opening and extension): control number/length of gaps; strong effect on motif continuity and global score.

- **Alignment mode** (global vs local): whether we force end-to-end alignment or focus on the best local region.

- **Treatment of end gaps / terminal penalties:** can change how N- or C-terminal tails are aligned or left unaligned.

3. If the aligned proteins were closer homologs, say, mouse and human, what type of matrix would you expect to work the best? Why?

For closely related proteins (mouse vs human viperin), a matrix for conserved / short evolutionary distance is preferred, such as BLOSUM80 or BLOSUM62. These matrices penalize unlikely substitutions more strongly and better capture fine-grained differences between highly similar sequences, while low-number matrices (BLOSUM45/30) are tuned for much more diverged pairs and would be unnecessarily permissive.

Task 3

Q1

Mode of the alignment	Identity	Similarity	Motif
Global (Needle)	70/430 (16.3%)	135/430 (31.4%)	KINFSGGEPF ...: . KIRITGGEPL
Local (Water)	68/319 (21.3%)	129/319 (40.4%)	KINFSGGEPF ...: . KIRITGGEPL

Q2

1. What is the main difference between the results of Needle (global alignment) and Water (local alignment)?

Needle (global) forces an alignment over the entire lengths of viperin and MoaA, so most of the alignment is weak and gap-rich, with only a small region around the SAM-binding motif looking reasonable.

Water (local) aligns only the best local region, effectively trimming unrelated N- and C-terminal parts. As a result, it highlights a short high-scoring block containing the SAM-binding-like motif and ignores the rest.

2. Do you think that the metrics (identity, similarity) are good in each case?

For the global alignment, 16% identity / 31% similarity over the whole length between proteins of different function is low and not very convincing as evidence of overall homology; much of it can be due to chance plus one shared motif.

For the local alignment, 21% identity / 40% similarity within a focused region is more indicative, especially when it coincides with the known SAM-binding motif.

3. Which mode is better to spot a functionally important motif?

Local alignment (Water) is better for detecting functionally important motifs shared by otherwise unrelated proteins. It concentrates on conserved blocks (like SAM-binding sites) and is not diluted by unrelated regions that global alignment is forced to align.

AI usage statement

I used OpenAI ChatGPT (GPT-5 Thinking) as an auxiliary tool while working on this assignment on 6 November 2025. Below are the specific interactions:

1. **Prompt:** "Read the presentation and the task in file `01_Pairwise_alignment_HW-1.docx`. Then check out my partial solution to the tasks in file `BioInf_01.pdf`. Check if everything is correct and help me fill out the rest. Especially theoretical questions."
2. **Prompt:** "Do AI usage note."
3. **Prompt:** "I think a proper statement should have the prompts listed."

The AI did **not** run sequence alignments for me and did **not** make independent experimental or numerical decisions. All sequence inputs, parameter choices in EMBOSS Needle/Water, interpretation of alignment outputs, and final conclusions were performed, checked, and approved by me.